Python Basics

Contents

Exercise 1 2
Exercise 2 2
Exercise 3 2
Exercise 4 2
Exercise 5 3
Exercise 6 3
Exercise 7 3
Exercise 8 4
Exercise 9 4
Exercise 10 5
Exercise 11 5
Exercise 12 5
Exercise 13 5
Exercise 14 5

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Exercise 1

```
roykoush@Koushanis-MacBook-Air Exercises % python3
Python 3.14.8 (main, Sep 30 2026, 17:55:09) [Clang 21.0.0
 ↪  (clang-2100.3.34.2)] on darwin
Type "help", "copyright", "credits" or "license" for more
 ↪  information.
>>> a = 1+2j
>>> b = 3+4j
>>> print(a,b)
(1+2j) (3+4j)
>>> r = a/b
>>> p = a*b.conjugate()
>>> print(r,p)
(0.44+0.08j) (11+2j)
```

Exercise 2

```
n = int(input('n: '))

#Left justified
for i in range(n+1):
    print(('*'*i).ljust(n))

#Right justified
for i in range(n+1):
    print(('*'*i).rjust(n))

#Centre justified
for i in range(n+1):
    print(('*'*(2*i-1)).center(2*n-1))
```

Exercise 3

```
n = 26
for i in range(27, 52):
    print(i%26, end=',')
```

Exercise 4

```
>>> import phyconsts
```

```
>>> help(phyconsts)

>>> print(dir(phyconsts))
['G', 'M_earth', 'M_sun', 'NA', 'R', 'R_earth',
 ↪  '__builtins__', '__cached__', '__doc__', '__file__',
 ↪  '__loader__', '__name__', '__package__', '__spec__', 'c',
 ↪  'e', 'epsilon0', 'g', 'giga', 'h', 'hbar', 'kB', 'kilo',
 ↪  'me', 'mega', 'micro', 'milli', 'mp', 'mu0', 'sigma']
>>>
```

Exercise 5

```
from turtle import *
mode('logo')
rt(30)
for i in range(3):
    fd(50); rt(120)done()
```

Exercise 6

```
import random
from turtle import *
L = [0, 90, 180, 270]
L2 = ['red', 'blue', 'green', 'purple']
reset()
mode('logo')
for i in range(20):
    color(random.choice(L2))
    rt(random.choice(L))
    fd(30)
    done()
```

Exercise 7

```
import turtle
turtle.mode('logo')

def draw_polygon(s, l):
    angle = 360 / s
    for i in range(s):
```

```
        turtle.fd(l)
        turtle.rt(angle)

draw_polygon(3, 100)
draw_polygon(5, 80)
turtle.done()
```

Exercise 8

```
n = int(input('Enter a number to get its multiplication
 ↪  table:'))
for i in range(1,101):
    print(n, 'x', i, '=', n*i)
```

Exercise 9

```
def print_butterfly(n):
    for i in range(1, n + 1):
        stars = "*" * i
        spaces = " " * (2 * (n - i))
        print(stars + spaces + stars)

    for i in range(n, 0, -1):
        stars = "*" * i
        spaces = " " * (2 * (n - i))
        print(stars + spaces + stars)

print_butterfly(5)
```

Output:

```
*         *
**       **
***     ***
****   ****
**********
**********
****   ****
***     ***
**       **
*         *
```

Exercise 10

```
from turtle import *
reset()
speed(0)
color('gold')
pensize(10)
for i in range(10):
    forward(250)
    right(144)done()
```

Exercise 11

```
st = "gcatgactattccttgac"ac = st.count('a')
cc = st.count('c')
gc_ = st.count('g')
tc = st.count('t')
print('The counts of a,c,g and t are respectively: ', ac, ',',
 ↪  cc, ',', gc_, ',', tc)
```

Exercise 12

```
st = "gcatgactattccttgac"
print(st[::-1])
```

Exercise 13

```
dna = "gcatgactattccttgac"
sfe = dna[-6]
print("6th base from the end:", sfe)
```

Exercise 14

```
dna = "gcatgactattccttgac"
pos = []
for i in range(len(dna)):
    if dna[i] == 'a':
        pos.append(i)
print("Positions of 'a' (0 index system):", pos)
```